Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/04/2005
PI Name:	Voorhees, Peter W. Ph.D.		
Project Title:	Coarsening in Solid-Liquid Mixtures (CSLM)-2		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Evanston	**State:**	IL
Zip Code:	60208-0893	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2000	**End Date:**	07/26/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:			
Flight Assignment:	NOTE (3/21/07): Project may be funded longer but current end date is 7/26/2007 (per N. Hall/ GRC)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2474		
Performance Goal No.:			
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:	Previous NASA sponsored work clearly showed that solid-liquid mixtures consisting of Sn-rich particles in a Pb-Sn eutectic liquid are ideal, and perhaps unique, systems in which to explore the dynamics of the Ostwald ripening process. The high coarsening rate in these systems permit accurate kinetic data to be obtained, and the thermophysical parameters necessary to make a comparison between theory and experiment are known. However, in a terrestrial environment experiments can be performed only at the relatively high volume fractions of solid where the presence of a solid skeletal structure prevents large-scale particle sedimentation. In these high volume fraction experiments, a comparison between theory and experiment shows that the solid-liquid mixtures are coarsening faster than predicted by an approximate theory for purely diffusional controlled Ostwald ripening. The objective of this project is to plan and perform a microgravity experiment on Ostwald ripening in solid-liquid mixtures. This experiment will serve two primary purposes: it will allow experiments to be performed which can be directly compared to heretofore untested theories for coarsening in systems with low volume fractions of solid, and it will eliminate conclusively convection of the liquid matrix and small-scale particle motion within the skeletal structure as possible sources for the disagreement observed between theory and experiment in the high volume fraction experiments. To perform coarsening experiments whose results can be compared directly with theory with no adjustable parameters. The spaceflight experiment will produce data which, for the first time, can be compared directly to theory with no adjustable parameters. This data will address the long standing controversy over the dependence of the coarsening rate of a two-phase system on the volume fraction of coarsening phase.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	CSLM-2 will make possible predictions of coarsening kinetics in alloys with high volume fractions of precipitates, such as alloys under consideration for high temperature exploration applications (i.e., nuclear propulsion, waste heat coolant loops). Results will be applied to materials design codes that will reduce cost, shorten development time, and improve performance over conventional materials design methods.
Task Progress:	We have examined three different groups of samples with volume fractions of solid of 20, 50 and 80%, respectively. The samples with volume fractions of 20 and 50% were processed in space. Using our serial sectioning machine we collected up to 400 two-dimensional microstructural images of each sample. These sections were then stacked into 3D microstructures that contained between 1000 and 10000 particles. These particles were then separated using the 3D watershed segmentation program that we designed. Using the data from the system with a volume fraction of 20%, we obtained the first three-dimensional measurements of the temporal evolution of the radial distribution function, particle size-particle size correlations and particle size distribution (PSD) of a system undergoing coarsening. The results are qualitatively similar to those obtained from an analysis of the microstructures using plane sections: the system has not reached the steady state coarsening regime. The next step is to compare the evolution of the measured PSD and spatial correlation functions with the predictions of a numerical simulation of coarsening. Since the experimental measurements provide the location and size of each particle, we can use this information from the earliest experiment to create the identical solid-liquid mixture on the computer. We can then allow the mixture to coarsen on the computer and can compare the evolution of the system predicted by theory to that measured experimentally with unprecedented accuracy. The three-dimensional microstructures of the 80 vol% and 50 vol% Sn samples were analyzed as well. The 3D PSD and the particle-particle contact distributions were determined. The 3D PSDs did not compare well to those predicted by particle coarsening theory. Surprisingly, there was a reasonable agreement with the grain size distributions predicted by grain growth theory. In addition, when the particle-particle contact distribution is normalized to the average number of particle contacts, the distribution was statistically invariant with volume fraction. At 50 vol% it is found that the number of contacts is proportional to the square of the average particle size, but this was not true for 80 vol%. This is attributed to the increased shape distortions of the particles that are present in the higher volume fraction samples. At higher volume fractions of coarsening phase, the particles are not spherical. Thus, it is necessary to use the volume-equivalent-sphere size in determining quantities such as the particle size distribution. We shall begin using methods that do not suffer from this simplification. We will be examining the evolution of the systems using the probability of finding a patch of surface with a given mean and Gaussian curvature. This more general approach will avoid the need to use the volume-equivalent-sphere size to characterize the microstructure. Along with this approach we are working on phase field methods to describe the evolution of such systems. Three-dimensional calculations using Cahn-Hilliard and Cahn-Allen equations are currently underway in an effort to understand the evolution of interfacial morphology in systems that are not a collection of spherical particles.
Bibliography Type:	Description: (Last Updated: 01/17/2024)
Articles in Peer-reviewed Journals	Rowenhorst DJ, Thornton K, Voorhees PW. "3-D Analysis of Particle Coarsening in High Volume Fraction Solid-Liquid Mixtures." in preparation, January 2005. , Jan-2005
Articles in Peer-reviewed Journals	Rowenhorst DJ, Voorhees PW. "Measurements of the grain boundary energy and anisotropy in tin." Metallurgical and Materials Transactions A. 2005 Aug;36(8):2127-35. http://dx.doi.org/10.1007/s11661-005-0333-7 , Aug-2005
Awards	Voorhees PW. "Fellow, American Physical Society, March 2005." Mar-2005
Presentation	Voorhees, P.W. "Coarsening in Solid-Liquid Mixtures: A Comparison Between Theory and Experiment" Department of Geology, Northwestern University Feb-2004